

Time Series Analysis report:

COVID-19 Global Daily New Cases

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1. Abstract

This report presents a comprehensive time-series analysis of the global daily new confirmed COVID-19 cases, using publicly available data from the Johns Hopkins CSSE repository. The workflow follows a standard pipeline: data acquisition, reshaping, aggregation, exploratory data analysis, stationarity testing, decomposition, and forecasting with naive, seasonal naive, and SARIMA models.

The analysis reveals strong non-stationarity, a clear weekly pattern, and multiple epidemic waves. While seasonal naive forecasting captures the weekly cycle, SARIMA provides the best short-term fit. However, classical models cannot fully anticipate structural breaks caused by new variants, policy changes, or reporting irregularities. The results are useful for short-term operational planning but should not be interpreted as long-term epidemiological predictions.

2. Introduction

2.1. Goal

The primary objective is to analyse the temporal evolution of the pandemic through a single global time series: the daily number of new confirmed cases. The analysis addresses three core questions:

- What temporal patterns (trend, seasonality, noise) are present in the data?
- How well do simple forecasting models perform?
- What are the main limitations of these models in practice?

2.2. Data source

All data are obtained from the Johns Hopkins University CSSE COVID-19 GitHub repository. The file `time_series_covid19_confirmed_global.csv` provides daily cumulative confirmed cases per country/province from 22 January 2020 to 9 March 2023.

2.3. Reproducibility

The entire analysis is implemented in a Jupyter notebook and can be reproduced by running the code with standard Python packages (`pandas`, `numpy`, `matplotlib`, `seaborn`, `statsmodels`). The data are fetched directly from the public URL, ensuring the same starting point for any replication.

3. Data Collection and Preprocessing

3.1. Confirmed cases data

The raw data are in a wide format (rows = geographic units, columns = dates). To make it suitable for time-series analysis, the notebook melts the table into a long format with `Date` and `Cumulative_Confirmed` columns. Dates are parsed to datetime objects.

3.2. Global aggregation

Daily global cumulative confirmed cases are obtained by summing over all countries for each date. The daily new cases are then computed as the first difference of the cumulative series:

$$\Delta C_t = C_t - C_{t-1}$$

where C_t is the cumulative confirmed cases on day t. This transformation removes the dominating upward trend and makes the series more tractable for modelling.

A practical limitation of this global aggregation should be noted. Different countries update their reports at different times, so summing across all countries can introduce reporting delays into the global series. This is accepted as inherent to the data rather than corrected for.

3.3. Why recovered and death data are not used

The notebook also loads the recovered and death time series. However, these series are incomplete and inconsistent (e.g., recovered data stop updating in mid-2021). The visual inspection (Figure 1) shows clear discontinuities, making them unreliable for the main analysis. Hence, the confirmed cases series is the only one used.

3.4. Recovered and death figure

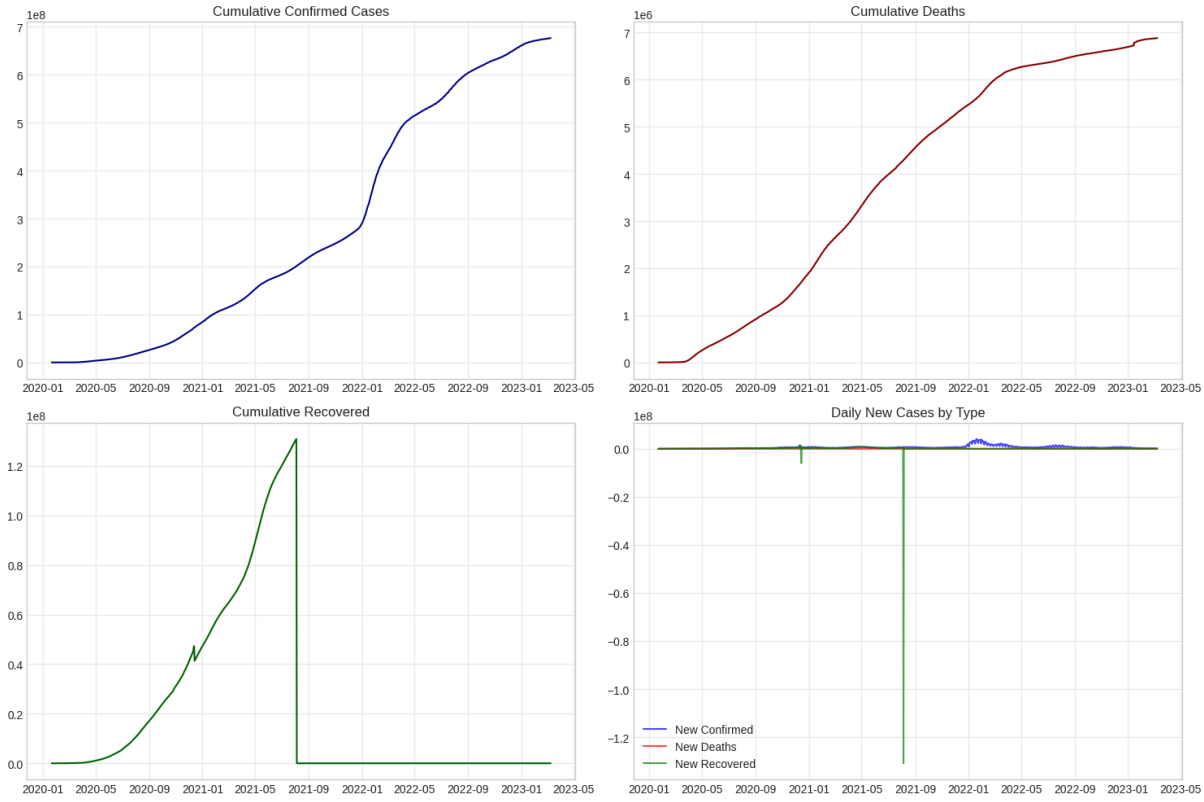


Figure 1: Recovered and death comparison

The figure demonstrates that the recovered and death curves are not consistently reported, reinforcing the decision to focus solely on confirmed cases.

4. Exploratory Data Analysis

4.1. Global shape of the series

The cumulative confirmed curve (Figure 3) exhibits a characteristic exponential growth in the early phase, later transitioning to a more linear growth pattern. The daily new cases series (Figure 4) is considerably noisier, showing multiple distinct peaks and troughs that correspond to real-world epidemic waves, but also to reporting artefacts (weekend dips, catch-up effects).

4.2. Interpretation of the peaks

The notebook identifies and interprets the main peaks:

- **Early 2020 (Jan–Apr)**: Global spread with cases plateauing at 0.1M, not forming a distinct peak. Mitigation stabilised counts without reducing them.
- **Mid-2020 (May–Oct)**: Linear growth from 0.2M to 0.7M driven by testing expansion. No clear peak or trough appeared.
- **Delta (Nov 2020 – mid-2021)**: Delta arrived at 0.8M. Repeated relaxation of measures prevented the usual post-wave drop, pushing cases steadily to 1.3M.
- **Omicron (late 2021)**: Omicron appeared at 2.1M. Its expected spike and crash were absorbed into the existing linear trend. Further relaxations removed the post-wave trough, so cases continued rising by 0.1M per month.
- **Decline (2022–2023)**: After the second Omicron wave, cases declined steadily through May 2023, driven by growing population immunity, seasonal changes, and stabilised public health responses.

These patterns are not purely biological; they are also influenced by testing capacity, reporting policies, and public health interventions.

4.3. Summary statistics

The daily new cases series has a mean of about 600,000 cases per day, with a similarly large standard deviation of around 600,000, which points to high day-to-day variability. The minimum is 0, which occurs in the early days of the pandemic, while the maximum exceeds 4 million cases in a single day.

4.4. Country comparison

The notebook extracts the top 10 countries by total cumulative cases and plots their daily new cases separately (Figure 5). Each country displays its own wave timing and amplitude. For instance, the USA and India have the largest absolute scales, while European countries often show earlier peaks. This heterogeneity justifies using the global aggregate for a univariate analysis, but also warns that any global model will smooth over important local dynamics.

4.5. Descriptive statistics

A summary table (mean, standard deviation, min, max, coefficient of variation) is computed for the top 10 countries. The coefficient of variation ($\text{CV} = \frac{\sigma}{\mu}$) is particularly high for countries like South Korea ($\text{CV} > 2$), indicating that their daily case counts are extremely volatile relative to their mean. The USA, while having the highest mean, has a lower CV (1.4), suggesting a more stable (though large) epidemic trajectory.

4.6. Correlation structure

A cross-correlation heatmap (Figure 6) shows moderate positive correlations among most countries, generally in the range of 0.2 to 0.6. This indicates that global waves are partially synchronised, but the correlations are far from perfect, reflecting different local conditions and responses. Interestingly, geographic proximity does not systematically translate into stronger correlation. France and Italy are the only clearly neighbouring pair that shows a strong link, while other neighbouring countries do not stand out. This suggests that factors such as long-distance travel and international connectivity may matter as much as physical proximity in synchronising waves between countries.

4.7. Multivariate analysis with functional principal component analysis

To go beyond pairwise correlations, the notebook applies Functional Principal Component Analysis (FPCA) to the daily new cases of the top 10 countries. FPCA extends classical PCA to functional data by treating each country's daily series as a smooth function, which makes it possible to identify the main shared modes of variation across countries rather than looking at each pair separately.

Each country's series is standardised (mean zero, unit standard deviation) before the decomposition, so that countries with very different scales can be compared fairly. The first three principal components are retained. The first two components are plotted against each other (Figure 7) to visualise how countries relate to one another in this reduced space, and the proportion of variance explained by each component is shown in a scree plot (Figure 8).

The first few components together explain a moderate share of the total variance, confirming that while there is a common structure across countries, a large part of the variation remains country-specific. This is consistent with the earlier observation that waves differ in timing and amplitude from one country to another.

4.8. Anomaly detection

Two complementary methods are used to identify unusual observations:

- **IQR rule**: An observation is flagged as an outlier if it lies outside $[Q_1 - 1.5 \cdot \text{IQR}, Q_3 + 1.5 \cdot \text{IQR}]$, where $\text{IQR} = Q_3 - Q_1$.
- **Z-score rule**: A point is considered anomalous if $|z| > 3$, with $z = \frac{x-\mu}{\sigma}$.

The IQR method detects about 8.1% of the points, while the Z-score method flags 2.9%. The overlap (2.9%) confirms that the most extreme events (like the Omicron peak) are detected by both. Many of these "anomalies" are not errors but genuine epidemic surges.

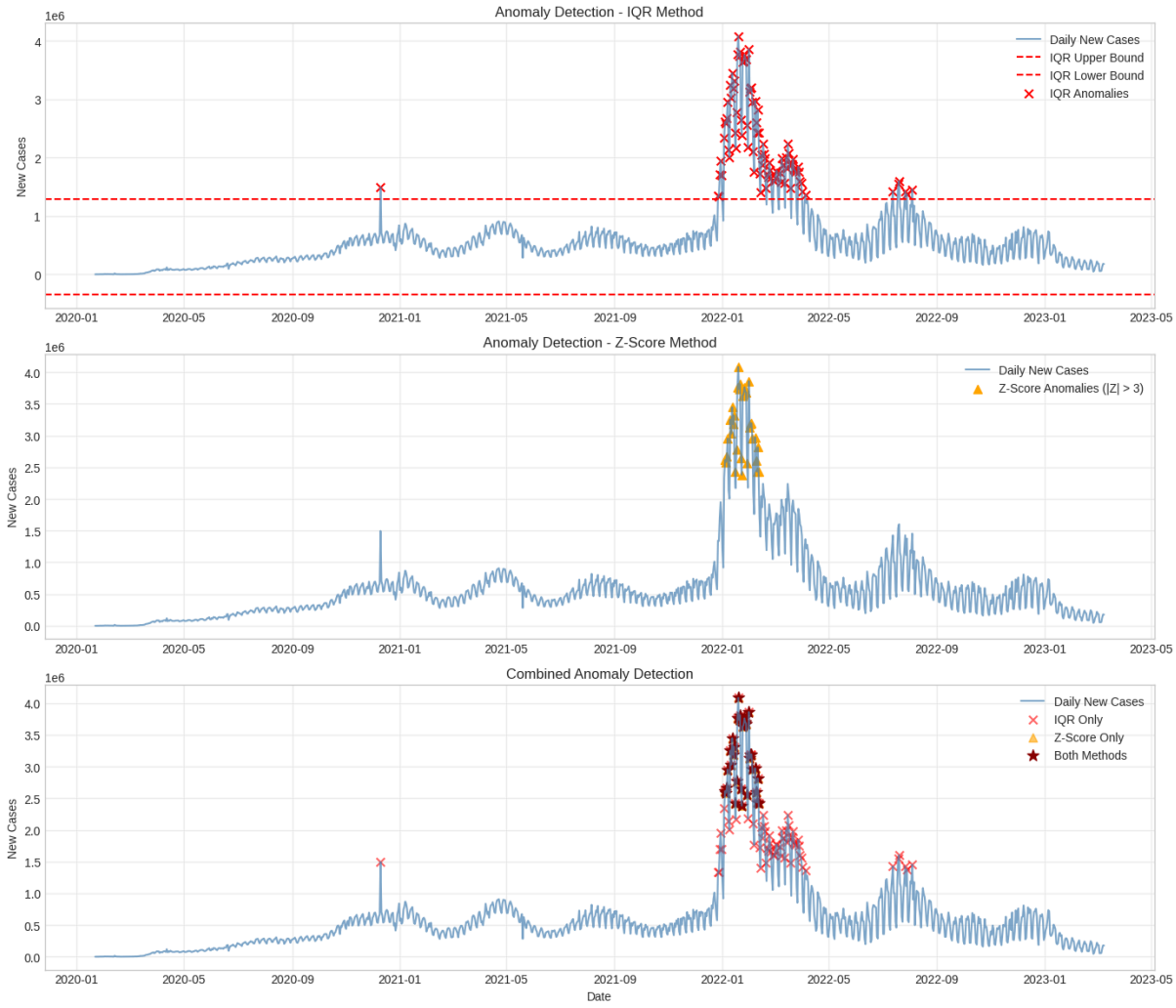


Figure 2: Anomaly detection summary

4.9. Figures for the global analysis

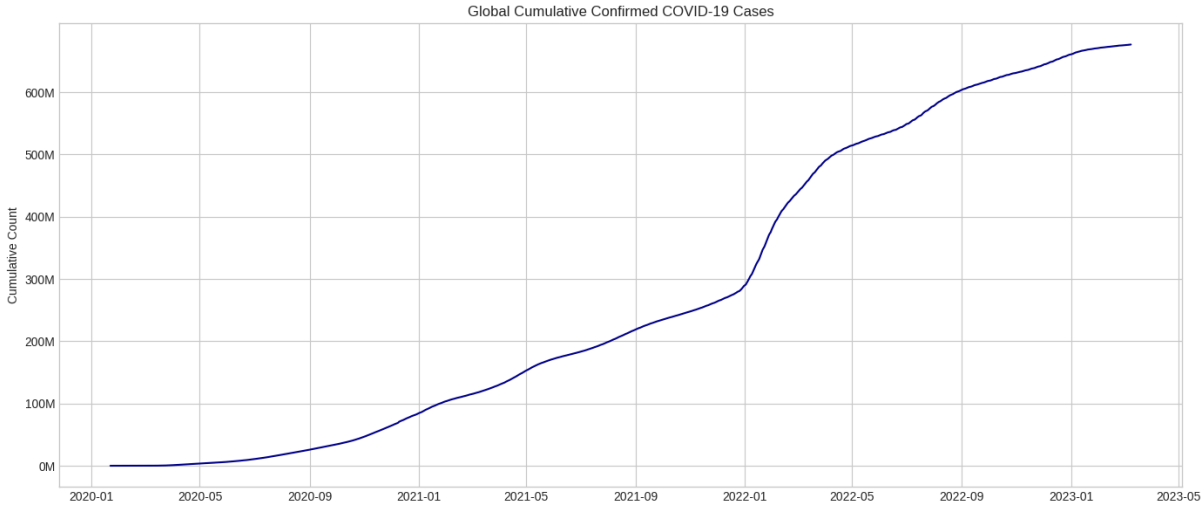


Figure 3: Global cumulative confirmed cases.

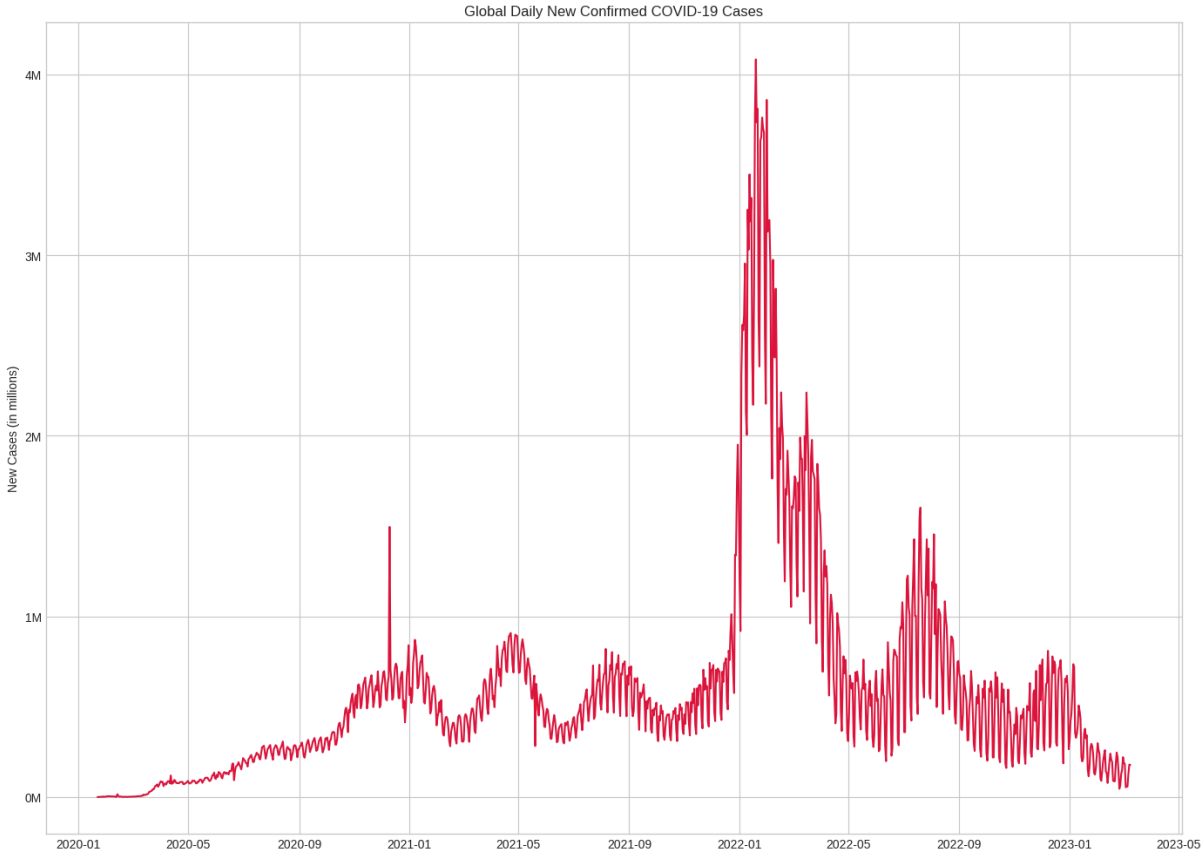


Figure 4: Global daily new confirmed cases.

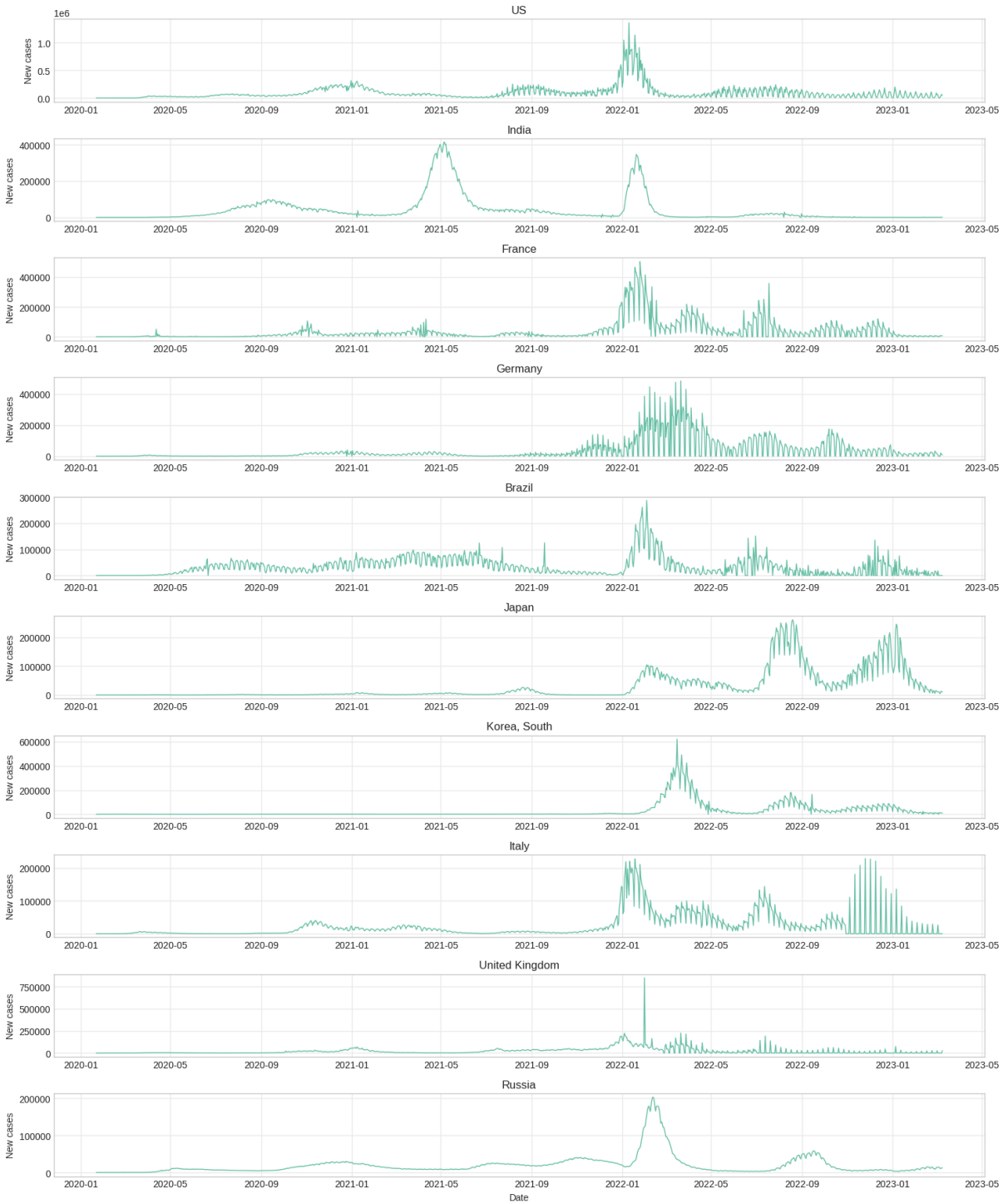


Figure 5: Daily new cases for the top 10 countries.

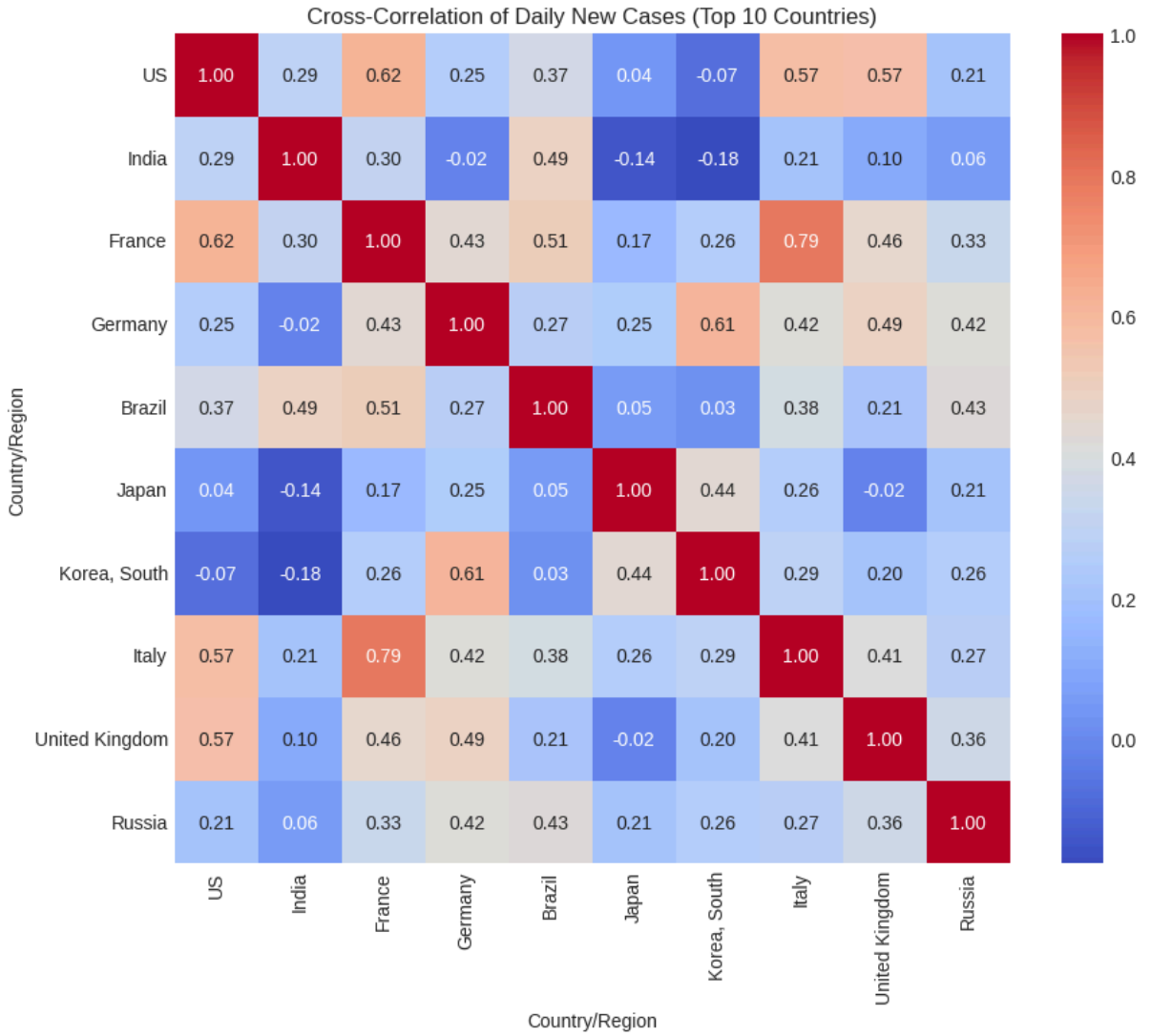


Figure 6: Cross-correlation heatmap for the top 10 countries.

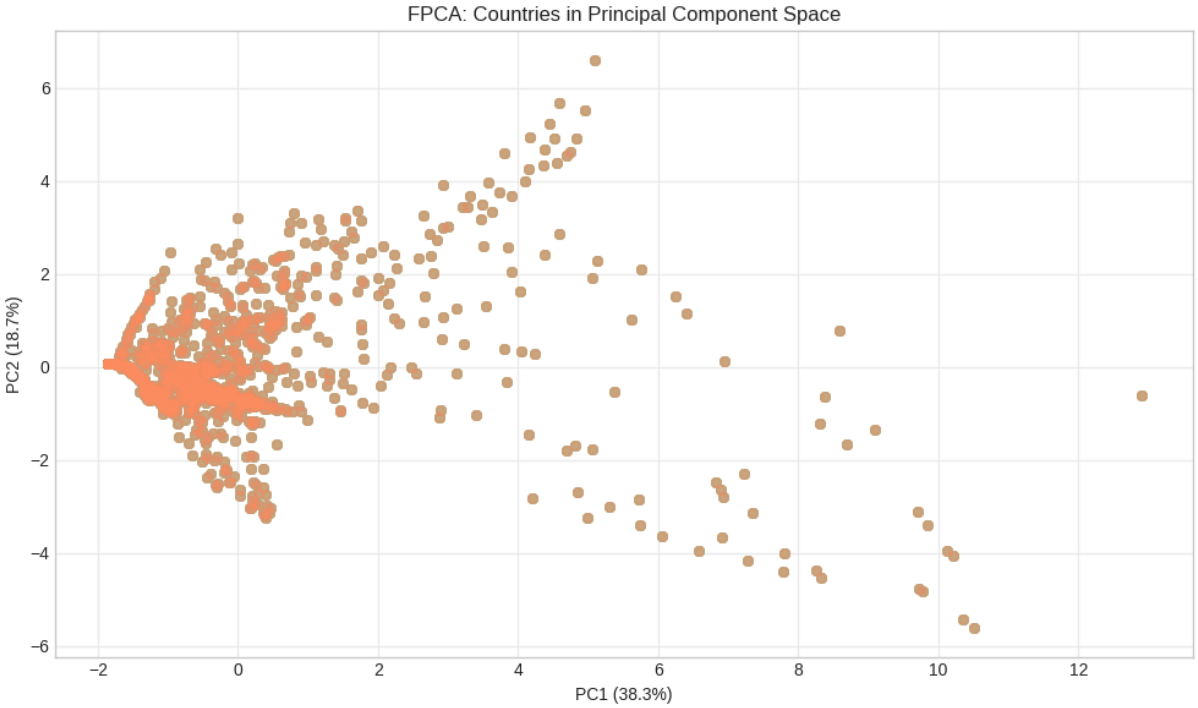


Figure 7: FPCA: countries in principal component space.

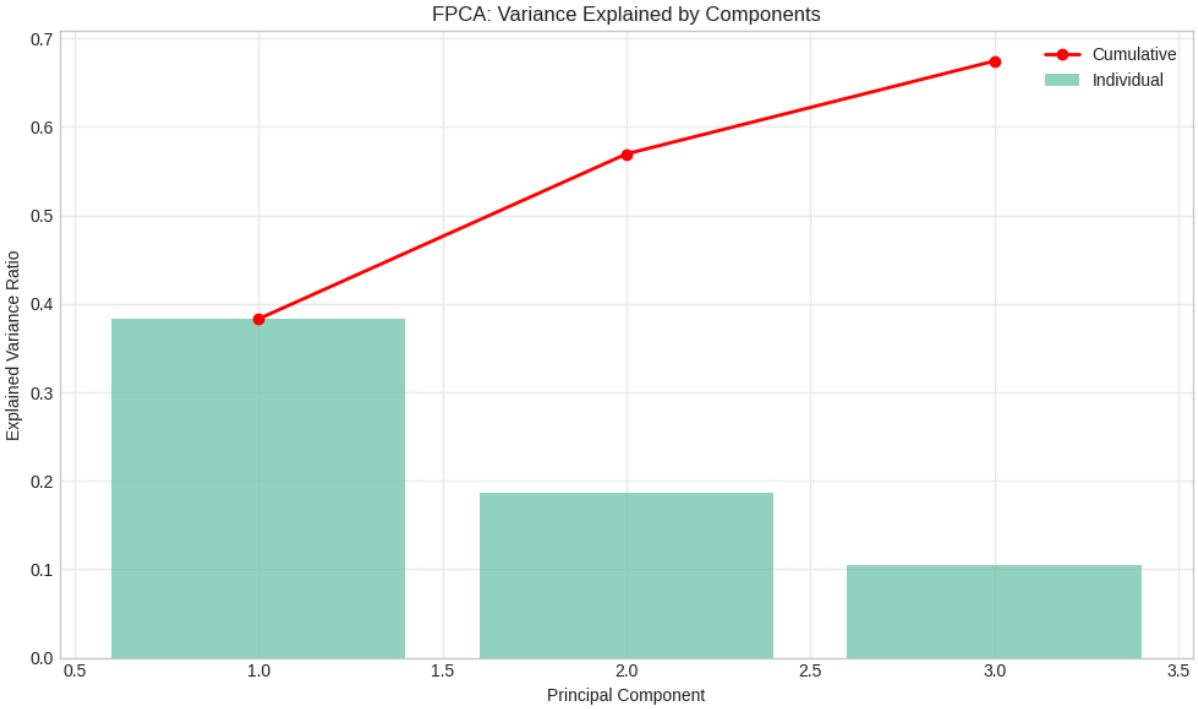


Figure 8: FPCA: variance explained by components.

5. Temporal Structure Analysis

5.1. Stationarity

The Augmented Dickey-Fuller (ADF) test and the Kwiatkowski-Phillips-Schmidt-Shin (KPSS) test are applied to assess stationarity.

- **ADF test**: Null hypothesis = unit root (non-stationary). The test fails to reject the null for the raw series, confirming non-stationarity.
- **KPSS test**: Null hypothesis = stationarity. The test rejects the null, again indicating non-stationarity.

After either first-differencing or seasonal differencing (lag 7), the tests indicate that the series becomes stationary, justifying the use of ARIMA-type models.

5.2. QQ plots

Normal QQ plots are used to visually assess the distribution of the data and later the residuals. The raw data show heavy tails, consistent with the presence of extreme outbreaks.

5.3. Seasonal decomposition

An additive decomposition (using STL) separates the global series into trend, seasonal, and residual components (Figure 9). The trend component captures the long-term evolution of the pandemic, while the seasonal component clearly reveals a weekly cycle (low reporting on weekends, catch-up early in the week). The residuals still contain autocorrelation, which is expected and is addressed by ARIMA modelling.

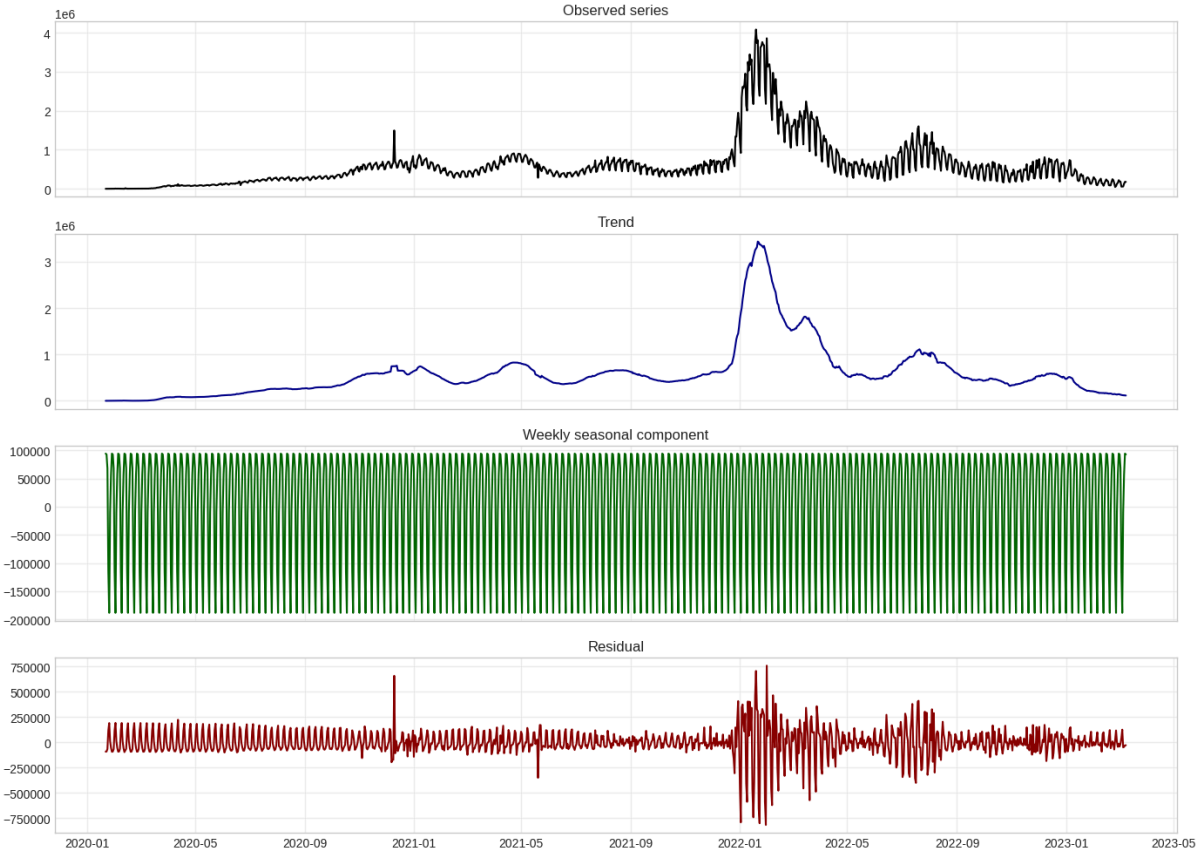


Figure 9: STL decomposition of the global daily new cases.

5.4. Calendar season effects

The notebook groups the data by meteorological season (winter, spring, summer, autumn) and compares distributions. Winter months tend to have higher medians and

more extreme peaks, which is consistent with the typical seasonality of respiratory viruses.

5.5. ACF and PACF

The autocorrelation function (ACF) of the raw series decays slowly, confirming non-stationarity. The partial autocorrelation function (PACF) of the differenced series shows significant spikes at lags 1 and 7, suggesting an AR(1) component and a seasonal AR component at lag 7. This guides the SARIMA order selection.

It is worth noting that the seasonal component obtained from decomposition is not meant to reproduce the trend. It only isolates the repeating weekly pattern once the long-term movement has been removed. This is why the trend, seasonal, and autocorrelation plots are inspected separately rather than expecting a single plot to explain everything at once.

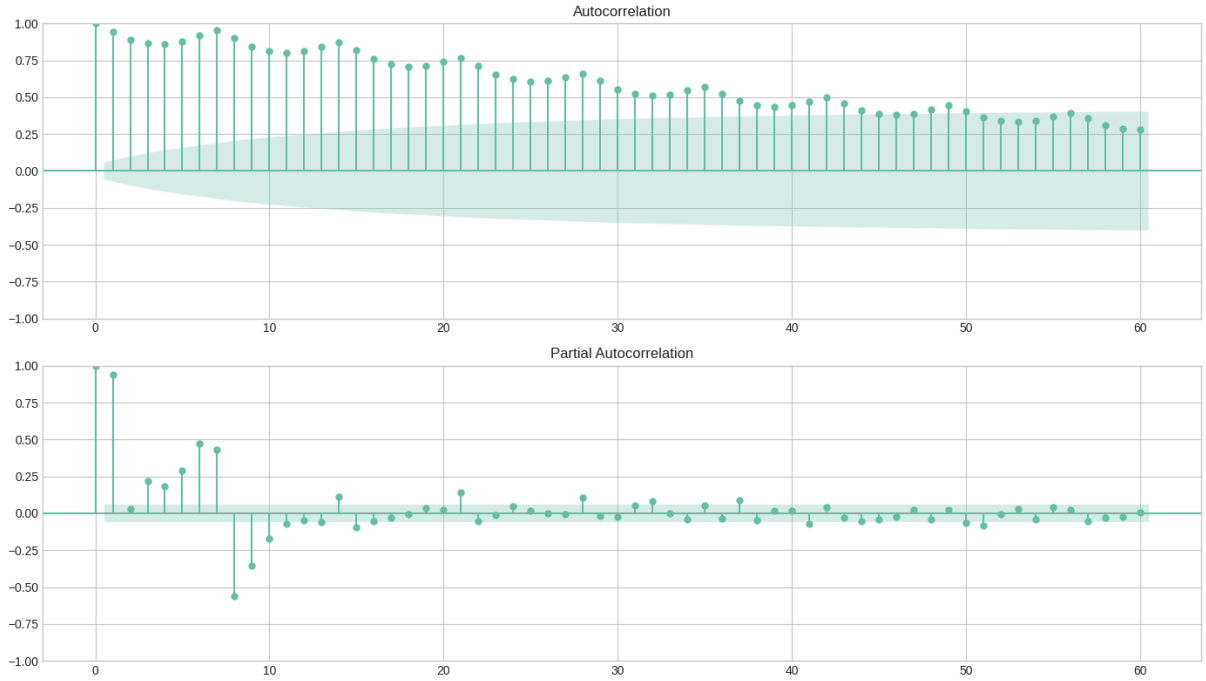


Figure 10: ACF and PACF of the differenced series.

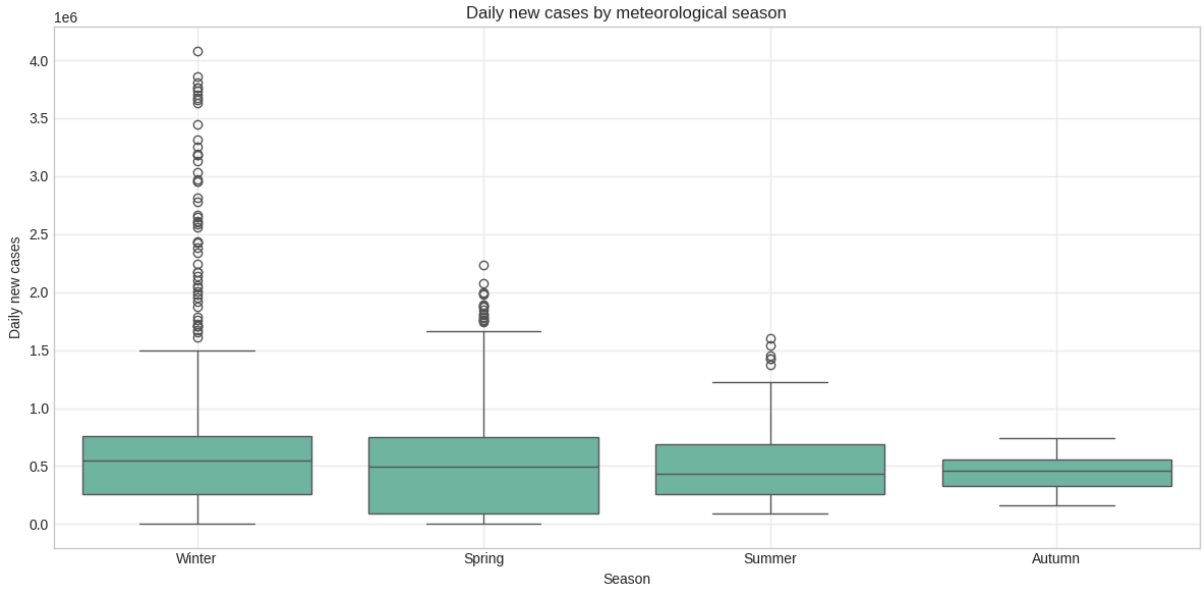


Figure 11: Daily new cases by meteorological season.

5.6. STL residual diagnostics

The STL residuals are further examined via time plot, histogram, QQ plot, and ACF (figures below). The residuals still show some autocorrelation, which the SARIMA model attempts to capture.

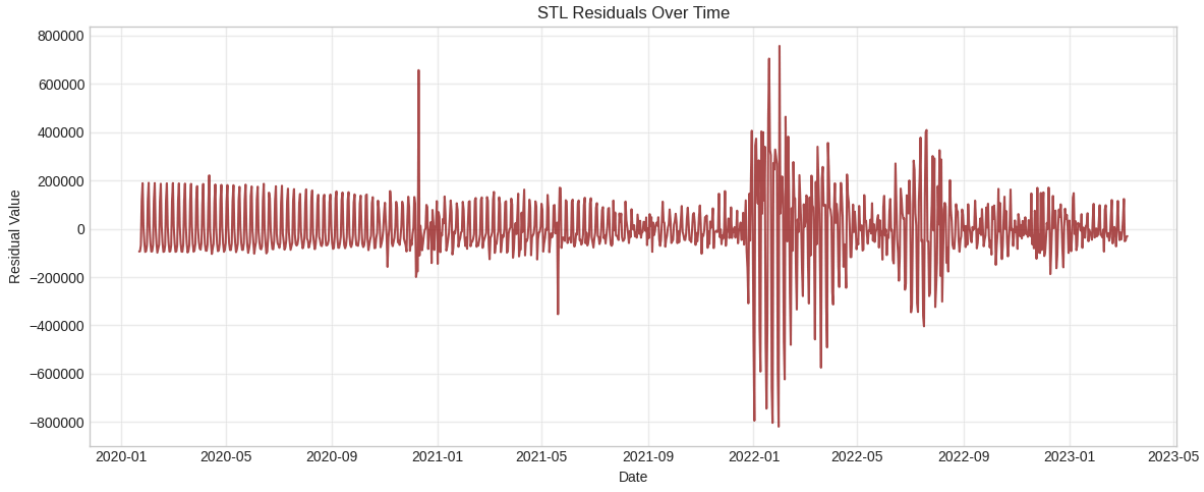


Figure 12: STL residuals over time

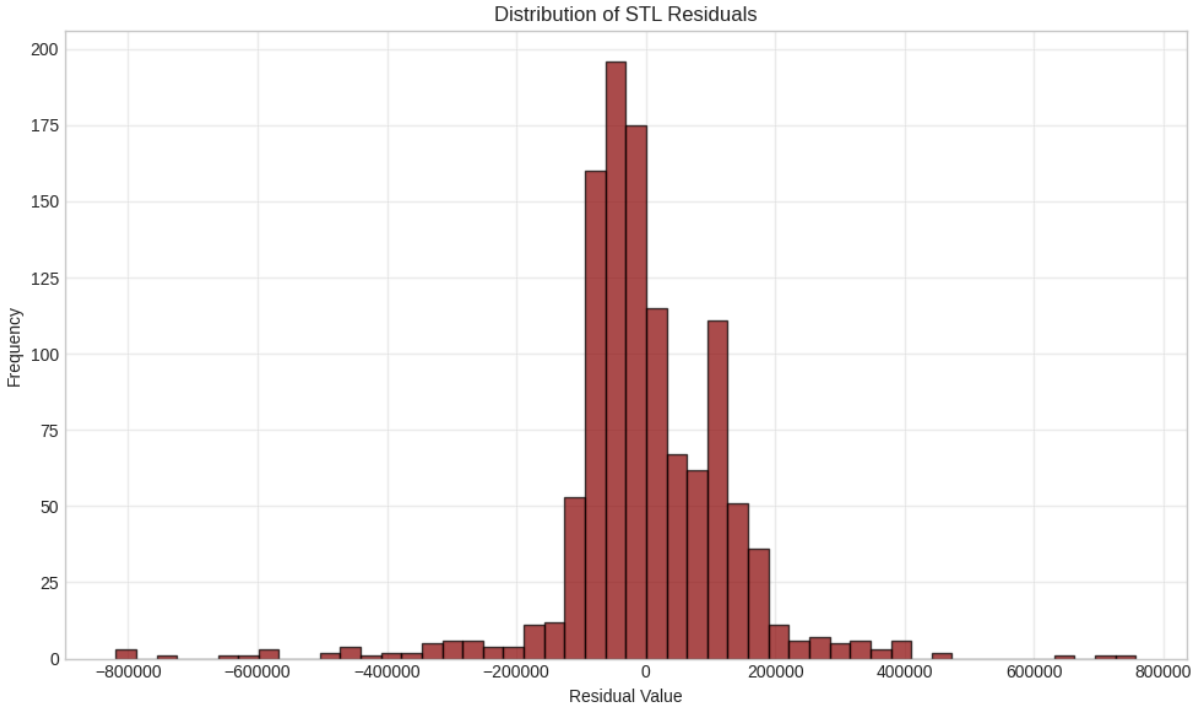


Figure 13: STL residual histogram

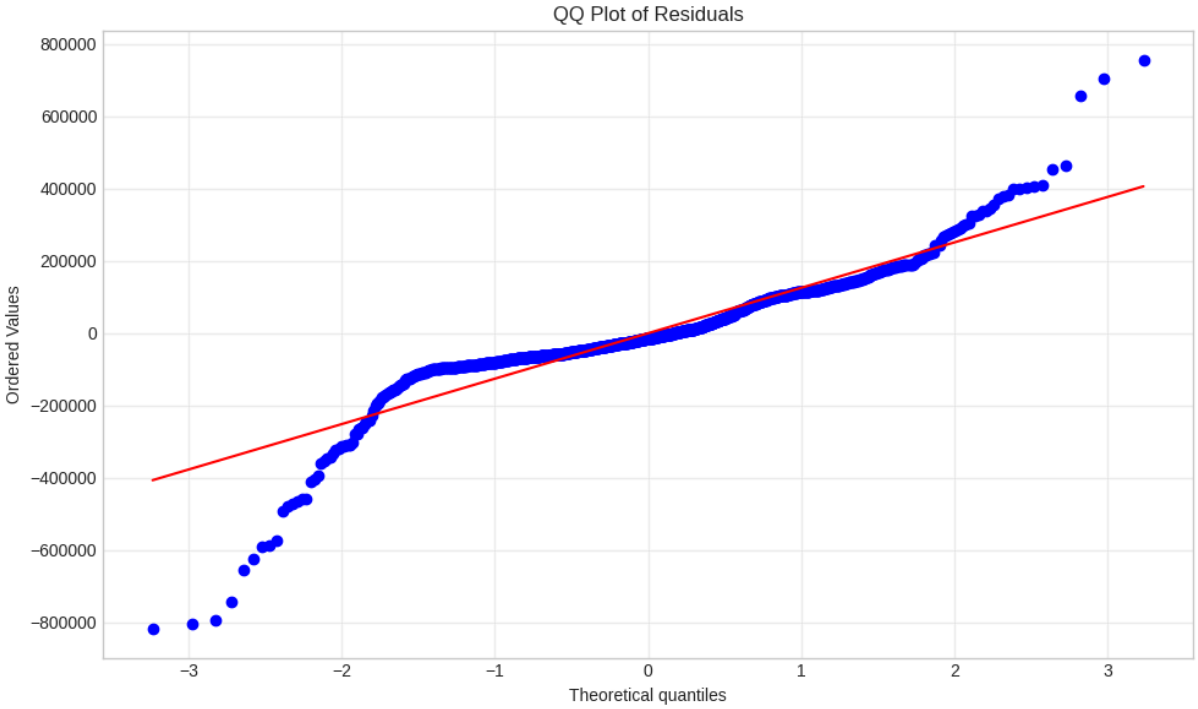


Figure 14: STL residual QQ plot

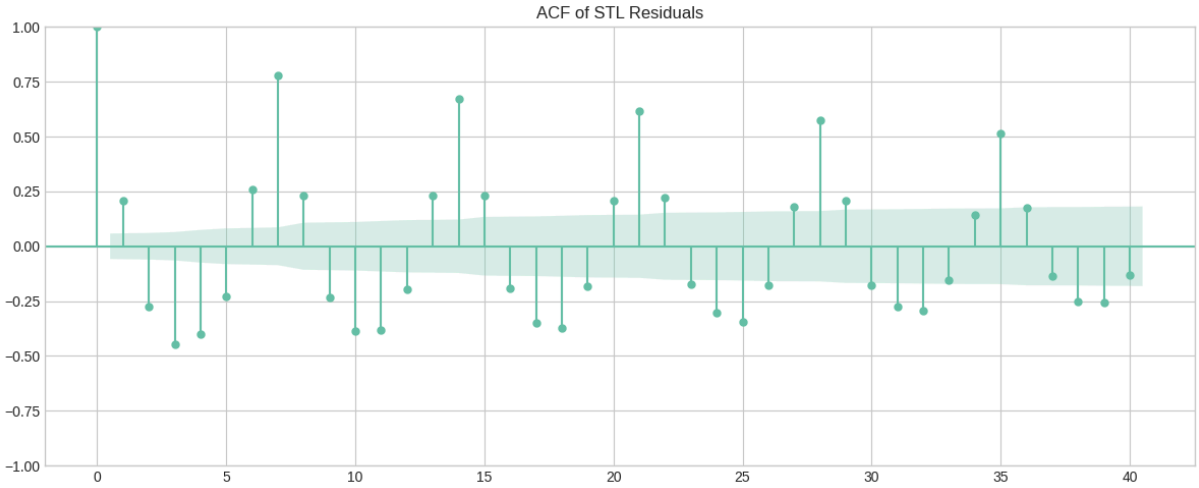


Figure 15: STL residual ACF

5.7. Temporal characteristics per country

To check whether the global patterns hold at country level, the seasonal decomposition and the ACF/PACF analysis are repeated for a small selection of countries (the United States, India, and Germany). The per-country trend, weekly seasonal component, and residual are shown in Figure 16, and the corresponding ACF and PACF plots are shown in Figure 17.

Each country shows the same overall structure as the global series, with a clear weekly seasonal cycle and a trend that follows its own local waves. The amplitude and timing of these components differ from country to country, which confirms once more that the global aggregate is a simplification that hides country-specific dynamics.

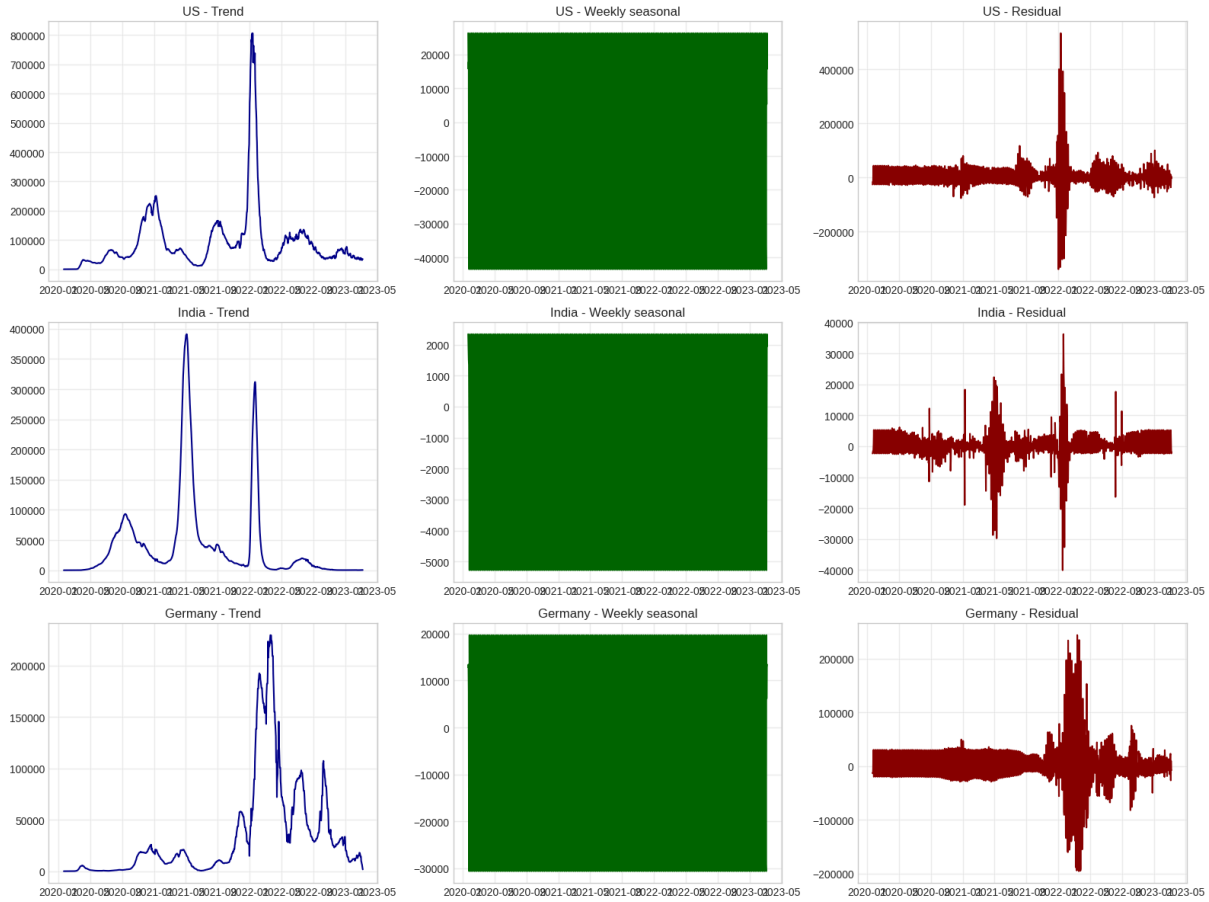


Figure 16: Trend, weekly seasonal component, and residual for the United States, India, and Germany.

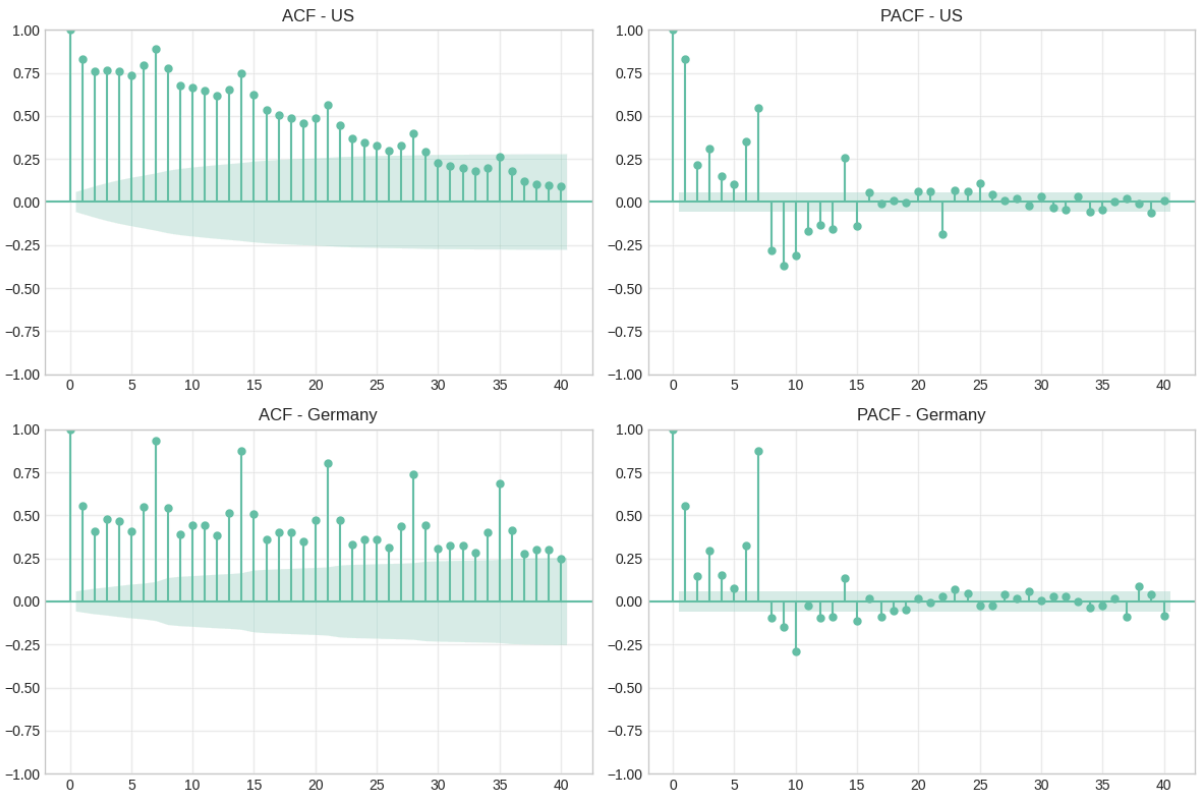


Figure 17: ACF and PACF for two selected countries.

6. Model Selection and Evaluation

6.1. Model choice

Three forecasting approaches are compared:

- **Naive**: $\hat{y}_{t+1} = y_t$
- **Seasonal naive**: $\hat{y}_{t+1} = y_{t-6}$ (i.e., the value from one week earlier)
- **SARIMA**: A Seasonal AutoRegressive Integrated Moving Average model, specified as SARIMA$(P, D, Q)_7$. The orders are chosen by minimising the Akaike Information Criterion (AIC) over a grid of plausible values.
- Hyperparameters choice: **SARIMA**$(1, 1, 1, 7)$ after a grid search on P, D, and Q over $[0; 2]$
- Season set to 7 according to seasonal decomposition

Main metrics used:

- Model tuning: **AIC** (described below)
- Model evaluation: **MAE**, **MSE**, **RMSE**

The AIC is defined as $\text{AIC} = -2 \log L + 2k$, where L is the likelihood and k is the number of parameters. Lower AIC indicates a better trade-off between fit and complexity.

These models are chosen because they are simple to interpret, well established for forecasting, and do not require the large amounts of data that more complex methods

such as deep learning would need. They also form a reasonable baseline for a series that is known to be non-stationary and seasonal.

The models rely on two simplifying assumptions. First, the trend and the weekly seasonality are assumed to stay roughly stable over the forecast horizon, which is not always true in practice. Second, the models assume that the near future will resemble the recent past, meaning that no major new intervention or variant changes the dynamics abruptly. This second assumption is a significant limitation, since it is precisely this kind of structural break that is hardest to predict but most consequential in an epidemic.

6.2. Train and test split

The data are split chronologically: the first $n - 30$ observations are used for training, and the last 30 days are held out for testing. This respects the temporal order and simulates a realistic forecasting scenario.

6.3. Forecasting results

The test-set performance is evaluated using Mean Absolute Error (MAE) and Root Mean Squared Error (RMSE). RMSE penalises large errors more heavily than MAE, which is relevant here because under or over predicting a peak in cases could have a serious practical impact. Both metrics are scale dependent, so they are only meaningful when comparing models on the same dataset rather than across different series.

The seasonal naive model outperforms the naive model, confirming the importance of the weekly pattern. The SARIMA model achieves the lowest errors, indicating that it captures additional short-term dynamics beyond the weekly cycle.

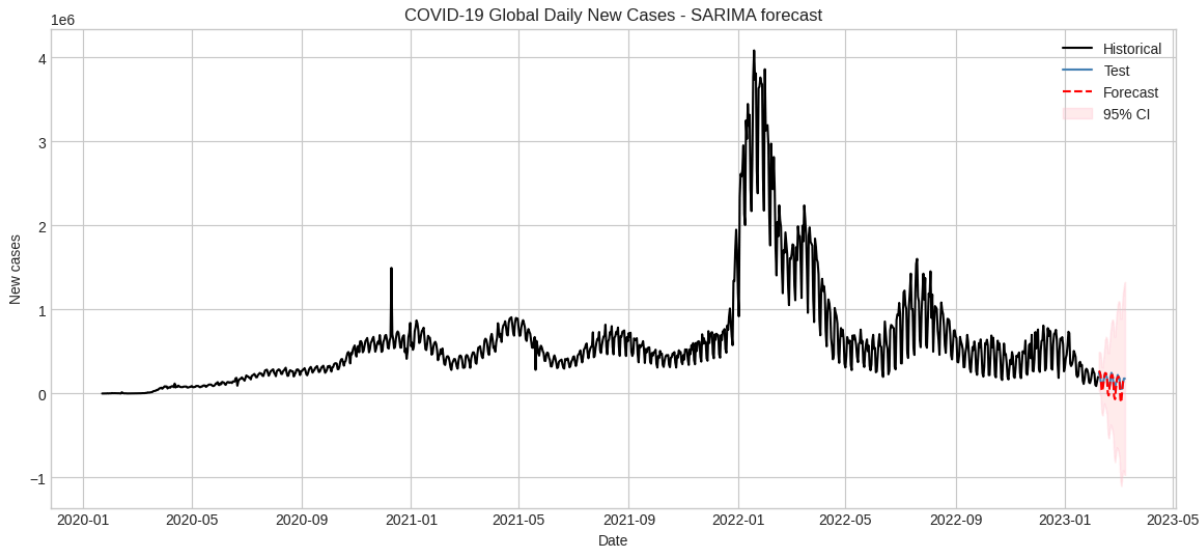


Figure 18: Comparison of forecasting methods on the test set.

6.4. Final forecast

The best-performing model (SARIMA) is retrained on the full dataset, and a 14-day ahead forecast is generated with 95% prediction intervals. The forecast shows a continuation of the recent downward trend, but the confidence intervals are wide, reflecting the inherent uncertainty in epidemic forecasting.

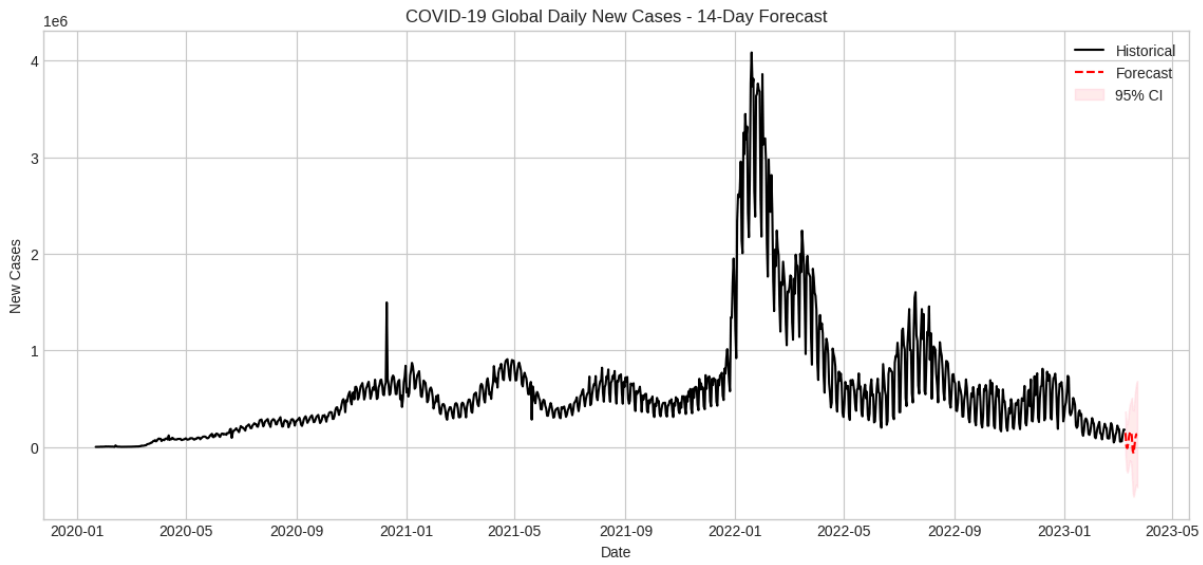


Figure 19: Final 14-day forecast with confidence interval.

6.5. Forecast figures

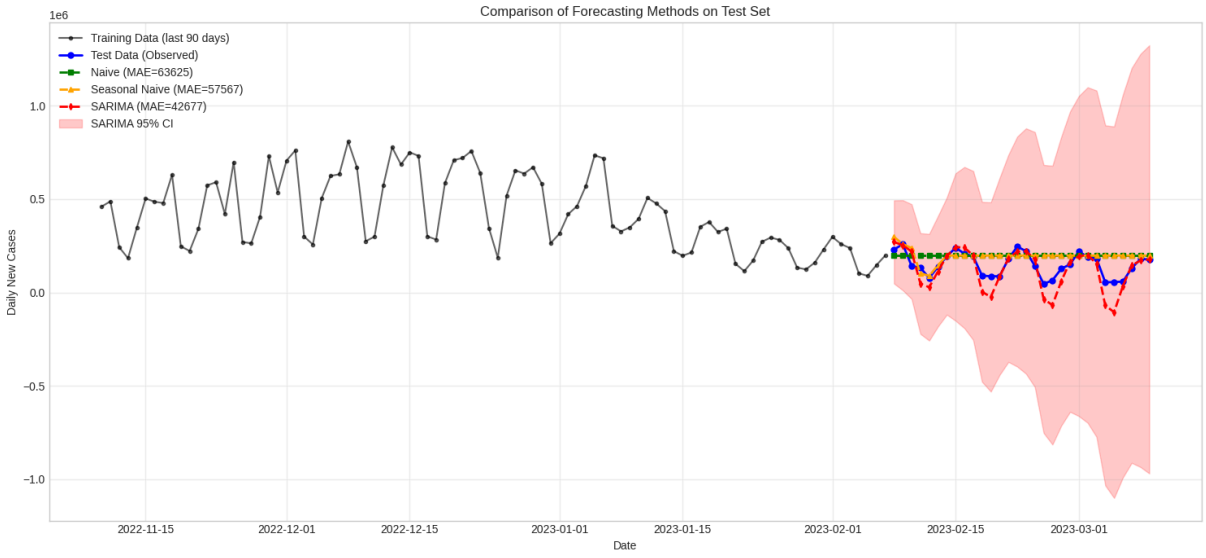


Figure 20: Forecasting methods comparison

7. Critical Discussion

7.1. What works

- The weekly seasonality is a robust feature that improves forecasting.
- Differencing and decomposition are effective for handling non-stationarity.
- SARIMA provides a flexible framework that captures both autoregressive and moving average components at the appropriate lags.

7.2. What does not work well

- Classical models cannot anticipate sudden structural changes (e.g., new variants, policy shifts).

- The assumption of stable weekly seasonality can break down when reporting practices or policies change over time.
- Global aggregation masks country-specific dynamics, which may be more predictable locally.
- The models ignore external factors such as vaccination coverage, testing capacity, and lockdown measures, all of which can strongly affect the trajectory of cases.
- Reporting delays and revisions introduce noise that is not captured by the model, and they affect the most recent data points the most, which are also the most important ones for forecasting.

7.3. Practical meaning

The forecasts are useful for short-term resource allocation but should be treated as conditional on the recent past. They are not a substitute for epidemiological modelling that incorporates transmission dynamics and intervention scenarios, such as SIR-type models. The results presented here are best understood as answering the question "what would happen if the recent trend continues", and should be complemented with expert judgement before being used for public health decisions.

8. Conclusion

The notebook delivers a complete time-series analysis of global COVID-19 daily new cases. It demonstrates the value of seasonal naive and SARIMA models for short-term forecasting, while also highlighting their limitations in a rapidly changing context. The analysis underscores the importance of careful data preprocessing, decomposition, and model evaluation in time-series projects.

9. Reproducibility Note

All code and data sources are publicly available. The analysis can be reproduced by running the provided Jupyter notebook with the required Python libraries. The report and notebook share the same methodology, ensuring traceability of all results and figures.